

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA
Denkleiers • Leading Minds • Dikgopolo tša Dihlalefi

COS700 Research Proposal

Diagnosing and Repairing XAI Fragility: SAM-Optimised Rules for Cancer Drug Response

Angelique Breedt u23542838
Bahiya Hoosen u22598546
Hawa Ibrahim u20418494
Melisa Ledwaba u19296950
Sakhile Manzi u26838002

Supervisors:
Prof. Nelishia Pillay
Dr. Nyathi Thambo

Abstract

Deep Neural Networks have achieved strong predictive performance in cancer pharmacogenomics, particularly for drug sensitivity prediction using high-dimensional genomic data. However, their black-box nature limits clinical adoption, as clinicians cannot verify whether model predictions are grounded in genuine oncogenic biology or learned statistical shortcuts, a phenomenon known as the Clever Hans effect. This study proposes and evaluates a RED XAI diagnostic pipeline that repositions explainability as an active model interrogation tool rather than a post-hoc justification mechanism. Using the Genomics of Drug Sensitivity in Cancer (GDSC) dataset and the biologically-informed DeepCCDS architecture, the pipeline applies Sharpness-Aware Minimisation (SAM) as a targeted intervention to steer the model toward flatter loss landscape minima. Symbolic if-then rules are extracted using a Genetic Algorithm and NSGA-II and evaluated across fidelity, comprehensibility, and stability via PROMETHEE II multi-criteria analysis, Wilcoxon signed-rank tests, and Gene Set Enrichment Analysis. The study hypothesises that SAM regularisation will produce rules that are statistically simpler, more faithful, and more strongly enriched in known oncogenic pathways than those extracted from an overfitted baseline.

Keywords

Explainable AI (XAI), Red XAI, Cancer, Precision Oncology, Genomics of Drug Sensitivity in Cancer, GDSC, Sharpness-Aware Minimisation (SAM), Generalisation Gap, Evolutionary Algorithms

1 Introduction

Deep Neural Networks (DNNs) have achieved state-of-the-art predictive accuracy in cancer genomics, particularly in tasks of predicting drug sensitivity using high-dimensional datasets such as the Genomics of Drug Sensitivity in Cancer (GDSC) database [4, 18]. Despite this predictive power, DNNs operate as black-box models whose internal logic is opaque to clinicians that utilise their predictive outputs to make life-determining decisions [3]. Consequently, their adoption in high-stakes clinical environments is hindered [3, 10] by what this study defines as the *Interpretability-Regularisation Gap*: the tension where highly accurate, complex black-box models rely on sharp, irregular loss landscapes that produce fragile, easily manipulated explanations, whereas intrinsically interpretable models lack the flexibility needed to capture highly non-linear biological relationships.

Explainable Artificial Intelligence (XAI) addresses this by extracting understandable symbolic rules from trained models [20], but the quality of these explanations is inseparable from the health of the underlying model. When a DNN overfits its training data, learning spurious correlations and dataset artefacts rather than genuine biological signal, a phenomenon known as the "Clever Hans" effect takes hold [2, 13, 15]. The dominant "Blue XAI" paradigm compounds these challenges by prioritising user-facing justifications over rigorous model auditing [6]. This project is instead grounded in "Red XAI", which fosters a culture of model interrogation and debugging [6], introducing a diagnostic pipeline that uses a model's generalisation gap as a measurable indicator of explanation fragility.

The remainder of this proposal is structured as follows. Section 2 formally states the research questions and hypothesis. Section 3 provides the theoretical foundation. Section 4 details the experimental methodology. Section 5 presents the project timeline.

Objectives

The objectives of this study are:

1. To establish a measurable overfitting baseline using the DeepCCDS architecture trained without regularisation on the GDSC dataset and to quantify the resulting generalisation gap.
2. To evaluate whether Sharpness-Aware Minimisation (SAM) regularisation reduces the generalisation gap and produces a flatter loss landscape relative to the Adam-optimised baseline.
3. To compare the fidelity, comprehensibility, and stability of symbolic if-then rules extracted by a Genetic Algorithm and NSGA-II from both the baseline and SAM-regularised models.
4. To determine whether rules extracted from the SAM-regularised model show statistically significant improvement in PROMETHEE II outranking scores over baseline rules using Wilcoxon signed-rank tests.
5. To assess whether SAM-optimised rules show greater enrichment in established KEGG oncogenic pathways than baseline rules, as verified through GSEA pathway enrichment analysis.

2 Problem Statement

Currently, there is no standardised diagnostic framework that uses a model's generalisation gap as a predictor of explanation quality. While regularisation is known to improve accuracy, research exploring how optimisers like Sharpness-Aware Minimisation (SAM) can be used to smooth the explanation landscape to produce rules that are faithful and comprehensible is limited [8, 16]. Without a "Red XAI" audit linking model health to rule quality, clinicians risk relying on explanations that may be mathematically convenient but biologically fraudulent [12].

The primary research question is as follows: to what extent can the generalisation gap serve as a diagnostic indicator of explanation fragility, and can SAM be used to improve the fidelity and comprehensibility of rules extracted using evolutionary algorithms?

This breaks down into four sub-research questions:

1. **Correlation**: Is there a statistically significant correlation between a model's generalisation gap and the rule explosion (complexity versus fidelity) observed in extracted symbolic rules?

2. **Regularisation Impact**: How does the transition from standard Adam to SAM-based optimisation affect the stability and sharpness of the model's error landscape, and the stability of its explanations?

3. **Multi-Criteria Validation**: To what degree do SAM-optimised rules outrank baseline rules when evaluated through the PROMETHEE II multi-criteria framework?

4. **Biological Alignment**: Do the rules extracted from a regularised model show higher enrichment in KEGG/Enrichr pathways compared to rules from an overfitted baseline?

The hypothesis is that applying SAM regularisation to a neural network trained on GDSC genomic data will reduce the generalisation gap by converging on flatter minima, resulting in symbolic rules that are statistically simpler, more accurately aligned to the black-box logic of the model (higher fidelity), and more consistently aligned with known biological pathways (higher biological validity) than those produced by an unregularised baseline [9, 11, 16, 20].

Scope and Limitations

This study is scoped to the following boundaries. The neural network architecture is fixed to DeepCCDS [17] and is not subject to architectural optimisation. The primary dataset is GDSC with LN IC_{50} as the drug response metric; the study does not extend to clinical patient data or real-time deployment contexts. The rule extraction pipeline operates on discretised tabular genomic features and does not address image-based or multi-modal data modalities. The regularisation intervention is limited to SAM as a replacement for the Adam optimiser; other regularisation strategies such as dropout or L2 weight decay are not evaluated as primary interventions. Statistical comparisons are conducted between the baseline and SAM-regularised conditions only; the study does not benchmark against all possible optimisers. Biological verification is limited to KEGG and Enrichr pathway databases as accessed through GSEApy; wet-lab experimental validation is outside the scope of this project.

3 Literature Study

3.1 Explainable AI (XAI)

Explainable Artificial Intelligence encompasses a suite of techniques designed to make the decision-making processes and predictions of complex AI models transparent and comprehensible to human users (verify citation) []. Interpretability and explainability are distinct concepts in AI [3]: interpretability (or transparency) describes the extent to which a model's internals naturally make sense to a human observer, while explainability is a process where a model provides specific details to clarify its functioning and decision-making. Based on this distinction, AI models can be classified into ante-hoc models (interpretable by design, e.g. decision trees, linear regression) and post-hoc techniques (applied to already-trained black-box models to extract information) [3].

3.1.1 Blue XAI vs Red XAI

[6] categorise XAI based on the goals of the explanations into two cultures: Blue XAI, which focuses on justifying decisions to build trust with end users, and Red XAI, which is oriented toward developers who use explanations as active diagnostic tools to explore, validate, and debug models. Table 1 summarises the differences.

Table 1: Comparison between Blue XAI and Red XAI, adapted from [6].

Feature	Blue XAI	Red XAI
Core Goal	Responsible, legal, trust, and ethical issues	Research on data, exploring models, and debugging
Target Audience	Lay users, customers, or patients	Power users, AI developers, and researchers
Explanation Type	Prioritises simplicity and comprehensibility	Prioritises faithfulness to the model and data
Primary Intent	Justify why a specific decision was made	Question the model, spot bugs, and extract knowledge

This project is rooted in the Red XAI culture, using evolutionary algorithms as a diagnostic and research tool rather than a mere justification mechanism [6, 20].

3.1.2 State-of-the-Art Post-Hoc XAI Methods

SHAP (SHapley Additive exPlanations) [WRITE: SHAP paragraph here]

LIME (Local Interpretable Model-Agnostic Explanations) [WRITE: LIME paragraph here]

Layer-wise Relevance Propagation (LRP) [WRITE: LRP paragraph here]

Integrated Gradients [WRITE: Integrated Gradients paragraph here]

Anchors [WRITE: Anchors paragraph here]
The limitations of these SOTA methods motivate the use of evolutionary rule extraction in this project. Unlike SHAP, LIME, LRP, and Integrated Gradients, which produce continuous feature-attribution scores that cannot be directly verified against biological pathways in a discrete, auditable form, evolutionary algorithms extract global symbolic if-then rules that can be explicitly evaluated for fidelity, comprehensibility, and biological validity [20]. Unlike Anchors, the GA and NSGA-II pipelines optimise rule quality as explicit objectives across the full GDSC dataset rather than locally per instance, and their outputs are directly compatible with the PROMETHEE II multi-criteria ranking framework [20]. Furthermore, and most critically, none of the SOTA methods described above can detect Clever Hans behaviour: [1] demonstrated that popular attribution methods including SHAP-style approaches produce visually plausible explanations even for models that have memorised noise, because their outputs are largely independent of model parameters. This project's Red XAI pipeline addresses this directly by auditing model health before any explanation is extracted [6]. Table 2 provides a structured comparison.

Table 2: Comparison of SOTA post-hoc XAI methods against the proposed EA rule extraction approach.

Method	Output	Scope	Auditable Rules	Fidelity Opt.	Bio. Verifiable
SHAP	Continuous scores	Local	No	No	Indirect
LIME	Continuous scores	Local	No	No	Indirect
LRP	Heatmap	Local	No	No	Indirect
Integrated Gradients	Continuous scores	Local	No	No	Indirect
Anchors	Rule (local)	Local	Partial	No	Indirect
GA + NSGA-II (Ours)	**Symbolic rules**	**Global**	**Yes**	**Yes**	**Direct (GSEA)**

3.2 Cancer Medicine and the Shift to Precision Pathology

Cancer is a mixed group of diseases that can be hematologically (blood) related or tumour related (National Cancer Institute, n.d.). They are all characterised by their uncontrolled cellular growth that is triggered by genetic mutations and epigenetic changes (environmental factors) alterations (National Cancer Institute, n.d.). Traditionally, treatment of cancer relied on generalised treatment protocols that were based on the anatomical site of the cancer. Modern protocols have shifted to what can be termed as precision medicine, which is aimed at providing customised individual patient care. Precision pathology in particular, aims to tailor treatment to patients based on the individual molecular profile of the patient's cancer. (Klauschen et al., 2024). Achieving this requires the integration of high-dimensional genomic data, such as gene expression profiles (transcriptomics), to predict how a specific cancer will respond to targeted therapies (Klauschen et al., 2024).

Precision pathology has evolved from manual microscopic inspection methods to AI-driven diagnostics. Machine learning models are now capable of analysing complex biological datasets to predict patient survival outcomes and drug sensitivities (Kovalev et al., 2021). A particular case is that of machine learning survival models. Machine learning survival models are predictive models designed to estimate the time-to-event outcomes, for example how long a patient is likely to live after diagnosis and or treatment (Kovalev et al., 2021). This is useful in determining the potency of a treatment plan based on the patient's characteristics (Kovalev et al., 2021).

3.2.1 Genomics and the Interpretability Gap

The precision of machine learning models in precision medicine is driven by massive datasets such as the Genomics of Drug Sensitivity in Cancer (GDSC). In genomic-based cancer research, datasets like (GDSC) provide thousands of gene expression variables. (Baptista et al., 2021; Yang et al., 2013). The GDSC dataset is a large publicly available database that is designed to help researchers discover therapeutic biomarkers (e.g. the expression levels of the EGFR gene) and predict how specific tumours will respond to various cancer treatments(revisit intro of paper) (Yang et al., 2013). It is practically, as well as financially, infeasible to test every candidate drug directly on human patients. Therefore, cancer cell lines are used as a proxy for real tumours (Yang et al., 2013). Cancer cell lines are cells that have been extracted and grown in a laboratory and still retain the genetic and molecular characteristics of the cancers they originate from (Yang et al., 2013). Drug response is measured in the form of IC50 values (Sakellaropoulos et al., 2019).

While DNNs excel at finding patterns in high-dimensional datasets such as GDSC, they largely operate as black boxes (Partin et al., 2023; Klauschen et al., 2024). To elaborate, current research has successfully used AI to identify biomarkers that correlate with treatment resistance or treatment efficacy (Partin et al., 2023; Klauschen et al., 2024; Baptista et al., 2020). Despite this success in predictive accuracy, a significant trust gap exists, which is that clinicians are not able to easily verify whether a model's prediction is based on genuine oncogenic pathways or statistical noise (Klaushen et al., 2024). Klauschen et al. (2024) argue that for AI to be integrated into clinical workflows, it must provide "explainable" outputs that align with established medical knowledge.

In XAI, unreliability of explanations is related to data corruption or noise (Klauschen et al., 2024).

In cancer genomics, small variations in the sample preparation can lead to batch effects which AI models learn as shortcuts to high accuracy or Clever Hans (Partin et al., 2023; Klauschen et al., 2024). Batch effects are non-biological, technical variations that are introduced into the data (Yu et al., 2024; Goh et al., 2022). Batch effects are not related to the variables of interest in the study and come about due to data being collected from different labs or machines, different assessment pipelines or differing experimental conditions over time (Yu et al., 2024).

Survival analysis research machine learning models shows that black box models can produce highly accurate predictions scores while hiding the underlying counterfactuals - specific changes in a patient's genetic expression can alter the clinical outcome or decision (Kovalev et al., 2023). Furthermore, if an XAI method identifies a "rule" for a cancer diagnosis that is mathematically sound but biologically incoherent (e.g., incorrectly linking a housekeeping gene to tumor suppression), it can lead to dangerous medical misinterpretations (Partin et al., 2023; Klauschen et al., 2024). This underscores the need for the rigorous "Audit and Verify" proposed in this research project to ensure that interpretability serves as a safeguard for clinical truth rather than just a justification for model output.

3.3 Model Sharpness and Generalisation

A well-documented limitation of deep learning models applied to high-dimensional biomedical data is their tendency to converge to sharp minima in the loss landscape, which is associated with poor generalisation and unstable learned representations. Li et al. (2018) demonstrate that neural networks can converge to either sharp or flat minima within the loss surface. Sharp minima are characterised by high sensitivity to parameter perturbations, meaning that small changes in weights can lead to large increases in loss. In contrast, flat minima exhibit smoother regions where the loss remains stable under perturbation. Li et al. show that flat minima are strongly associated with improved generalisation performance, while sharp minima are linked to overfitting and poor transferability to unseen data. This phenomenon is directly reflected in the generalisation gap, where a large discrepancy between training and test performance indicates that the model is memorising noise rather than learning meaningful biological structure (Keskar et al., 2016).

The relationship between loss landscape geometry and explanation quality is less established. Dombrowski et al. (2022) demonstrate that explanation methods are sensitive to the geometry of the model's decision boundary, and that smoother boundaries produce more stable attributions. Springer et al. (2024) further show that optimisers targeting flat minima improve the quality and balance of learned feature representations. However, whether this geometric improvement translates into measurable gains in the quality of symbolic rules extracted from a model, particularly in the pharmacogenomic domain, has not been empirically demonstrated. We argue that improving generalisation through landscape-aware regularisation is likely to improve the fidelity and comprehensibility of extracted rules, but this remains the central untested claim of this project, formalised as the hypothesis in Section 2 and addressed as a gap in Section 3.7.

3.3.1 Sharpness-Aware Minimisation (SAM)

SAM has been proposed as a mechanism for improving model generalisation by explicitly guiding optimisation toward flat minima. Unlike standard gradient descent, which may converge to sharp and unstable regions of the loss landscape, SAM introduces a local adversarial perturbation step that evaluates model performance within a neighbourhood of the current parameters, optimising for solutions that remain robust under small perturbations (Foret et al., 2021). This encourages convergence toward flatter minima, which have been strongly associated with improved generalisation and stability in deep neural networks (Li et al., 2018; Keskar et al., 2016). Critically, Springer et al. (2024) demonstrate that this geometric effect produces more balanced feature representations than conventional regularisers such as L2 weight decay or dropout, which reduce overfitting through penalty or stochastic noise but do not explicitly target loss landscape geometry or the distribution of feature importance across the model. This property is particularly relevant for downstream rule extraction, where rules derived from a model with balanced feature weights are expected to be shorter and more generalisable than those from a model that has concentrated weight on spurious correlates. As a result, SAM improves both predictive robustness and the consistency of explanations produced by downstream XAI methods.
The use of SAM is particularly important for the XAI pipeline defined in this project. As highlighted

by Guidotti et al. (2020), the reliability of local explanation methods is highly dependent on the stability of the underlying model. If a model is trained in a sharp region of the loss landscape, even minor perturbations in input or parameters can produce substantially different explanations, reducing trust in rule-based outputs. By enforcing flatter minima through SAM, the model is expected to yield extracted rules that remain consistent, interpretable, and aligned with underlying biological signals rather than statistical noise.

3.4 Artificial Neural Networks

Artificial Neural Networks (ANNs), and specifically feedforward Multi-Layer Perceptrons (MLPs), have served as a standard baseline architecture for regression tasks on tabular biomedical data, trained via backpropagation to minimise a defined loss function such as Mean Squared Error for continuous targets like drug sensitivity IC_{50} (Goodfellow et al., 2016; Arrieta et al., 2020). Despite their widespread application, standard MLPs present a well-documented limitation when applied to high-dimensional genomic data: they are inherently prone to overfitting, particularly when the number of input features vastly outnumbers the number of biological response samples (Partin et al., 2023). The GDSC dataset, after preprocessing via the approach established by Alipour (2024), retains thousands of gene expression features across cancer cell lines and drug combinations. When an MLP is trained without adequate regularisation, it tends to memorise training-specific noise rather than capture the true underlying biological relationships between gene expression profiles and drug response. This manifests as a high generalisation gap, a large difference between training accuracy and validation accuracy (Ye & Farnia, 2025), and constitutes precisely the "Clever Hans" phenomenon that motivates this research project (Samek et al., 2017; Anders et al., 2022). Pahde et al. (2023) demonstrate through their Reveal to Revise (R2R) framework that state-of-the-art deep neural networks frequently learn spurious correlations embedded in training data, including medically irrelevant artefacts, and that post-hoc XAI methods applied to such overfitted models produce explanations that are visually plausible but functionally unreliable. Although R2R was developed in an image classification context, the underlying mechanism, a model achieving high training performance by exploiting dataset-specific shortcuts rather than genuine signal, applies equally to tabular genomic settings, as documented by Partin et al. (2023) in the drug response prediction domain. For the purposes of this project, an intentionally overfitted baseline model serves as a diagnostic tool, its wide generalisation gap becomes evidence of fragile explanations, which the SAM regularisation intervention in Stage 2 is designed to correct.

3.4.1 Deep Learning for Cancer Drug Sensitivity Prediction

Beyond the standard MLP baseline, recent work has demonstrated that domain-informed deep learning architectures significantly outperform generic feedforward networks on drug sensitivity prediction tasks using GDSC data (Partin et al., 2023; Wu et al., 2025). The key limitation of a plain MLP is that it treats all gene expression features as equally independent, ignoring the rich biological structure known to govern cancer drug response – specifically, the organisation of genes into signalling pathways that collectively regulate cancer cell development, proliferation, and resistance mechanisms.
DeepCCDS (Wu et al., 2025) is a deep learning framework designed to predict cancer cell drug sensitivity by characterising cancer driver signals.
To overcome the limitations of standard neural networks, it replaces high-dimensional gene expression data with a prior knowledge network consisting of 38 biologically meaningful signalling pathways (Wu et al., 2025) . Alongside these pathway features, the architecture utilises self-supervised autoencoders to compress gene mutations and drug structures into low-dimensional embeddings . As demonstrated via t-SNE visualisations, these refined representations effectively separate drug-sensitive from drug-insensitive cell populations, confirming the model captures true biological structure rather than statistical noise (Wu et al., 2025). DeepCCDS was trained on 319,543 drug–cell line pairs from the GDSC database, demonstrating superior predictive performance over state-of-the-art methods (Wu et al., 2025). Crucially, its clinical applicability was validated on solid tumour patient samples from the TCGA cohort, where predicted LN IC_{50} values successfully distinguished between responsive and non-responsive patient–drug combinations (Wu et al., 2025). The interpretability properties of DeepCCDS are directly relevant to the goals of this project. By integrating prior knowledge about cancer driver signals, DeepCCDS exhibits powerful feature representation capabilities and built-in interpretability.The pathway-level structure of the architecture makes it possible to trace which biological mechanisms the model weights most heavily when predicting drug response. This is critically important in our RED XAI framework:

when evolutionary algorithms extract symbolic rules from the model, genes appearing in those rules can be cross-referenced against the same biological pathways that structured the model's internal representations, providing a direct route to biological verification via GSEA pathway enrichment analysis (Section 3.6.4).

3.4.2 The Tendency of Deep Models to Overfit Genomic Data and the Clever Hans Problem

As introduced in Section 1 and contextualised for genomic data in Section 3.2.1, the Clever Hans problem represents a direct threat to explanation quality in overfitted models. In the genomic setting specifically, Partin et al. (2023) document that the curse of dimensionality, thousands of gene expression features measured across a comparatively limited number of cell line–drug combinations, creates conditions under which unregularised models routinely converge on sharp loss landscape minima and memorise training-specific noise rather than genuine biological signal (Li et al., 2018).
The methodological precedent most relevant to this project is the Reveal to Revise (R2R) framework of Pahde et al. (2023), which proposes an iterative cycle of bias identification, artefact localisation, model correction, and re-evaluation. R2R demonstrates that when XAI methods such as LRP are applied to overfitted models, the resulting explanations attribute high relevance to artefactual features, producing outputs that are mathematically consistent with the model's behaviour but medically invalid (Pahde et al., 2023; Klauschen et al., 2024). This audit-and-fix philosophy directly anticipates the RED XAI diagnostic pipeline this project operationalises through SAM regularisation and evolutionary rule extraction, with the key distinction that this project applies a regularisation intervention at the optimisation stage rather than correcting artefacts post hoc. In this project, the baseline configuration of DeepCCDS (Wu et al., 2025), trained without regularisation to intentionally exhibit overfitting, serves as the Fragility Baseline against which the SAM-regularised model is compared in Stage 2. The generalisation gap at this stage is the primary diagnostic indicator of explanation fragility, as formalised in Section 2.

3.5 Evolutionary Algorithms

Evolutionary algorithms (EAs) are a class of stochastic search and metaheuristic optimisation techniques inspired by the biological processes of natural evolution and selection. Unlike traditional algorithms that iterate on a single solution, EAs maintain a population of individuals, where each individual represents a potential solution to a problem (Yedjour et al., 2024). The population evolves over generations using three primary operators: selection, crossover, and mutation.

Rule extraction is a post-hoc explainability technique used to identify the underlying decision-making logic of opaque black-box models such as ANNs (Ghasemi et al., 2024; Yedjour et al., 2024). It is fundamentally a multi-objective optimisation problem requiring a simultaneous balance between fidelity to the black-box model, predictive accuracy on unseen data, and rule comprehensibility (Yedjour et al., 2024). Deterministic rule extraction methods such as decision trees impose fixed inductive biases on the search, for example, axis-aligned splits that limit depth and greedy coverage strategies, which constrain their ability to navigate the combinatorial complexity of high-dimensional genomic feature spaces or to optimise competing objectives simultaneously (Arrieta et al., 2020; Yedjour et al., 2024). EAs are well suited to this problem precisely because they maintain a diverse population of candidate rule sets, can be configured with flexible multi-objective fitness functions that reflect fidelity, accuracy, and comprehensibility simultaneously, and are capable of escaping local optima through mutation and crossover operators (Yedjour et al., 2024). Among EA variants, Genetic Algorithms and NSGA-II are selected for this project: GAs provide a single-objective baseline for rule extraction quality, while NSGA-II explicitly evolves a Pareto front of non-dominated rule sets, enabling a structured trade-off analysis between fidelity and comprehensibility that single-objective methods cannot produce (Yedjour et al., 2024).

3.5.1 Genetic Algorithms

Genetic Algorithms were first introduced by John Holland in the 1970s (Holland, 1975) and represent a well- established metaheuristic optimisation technique inspired by natural evolution and selection (Yedjour et al., 2024). They are designed to solve complex problems where traditional methods struggle due to high dimensionality or non-linearity of the search space (Yedjour et al., 2024).
A genetic algorithm operates through an iterative process to evolve a population of potential solutions

toward an optimal result. Each individual in the population represents a candidate rule set encoded as a chromosome, and is assessed using a fitness function that determines how well it solves the target problem (Yedjour et al., 2024). The best-performing individuals are selected as parents for the next generation, commonly through tournament selection or roulette wheel selection (Yedjour et al., 2024). New offspring are produced through two biological operators: crossover, which combines genetic information from two parent chromosomes to create new individuals, and mutation, which introduces random changes to maintain population diversity and prevent convergence on local rather than global optima (Yedjour et al., 2024).

3.5.2 Application in Rule Extraction and XAI

Genetic algorithms serve as a powerful tool for rule extraction from black-box models like Artificial Neural Networks. This process involves identifying the decision-making logic of a complex network and converting its non-linear relationships into a set of interpretable "if-then" rules (Yedjour et al.2024).When used for this purpose, the GA evaluates "successful" rule sets based on three primary criteria such as fidelity, accuracy, and comprehensibility. Fidelity covers how closely the extracted rules mimic the decision-making process of the original ANN. Accuracy is how well the rules perform when classifying new, unseen data and lastly Comprehensibility is whether the rule set is simple and manageable enough to be easily understood by human experts (Yedjour et al., 2024).

3.5.3 Non-Dominated Sorting Genetic Algorithm II (NSGA-II)

While standard Genetic Algorithms usually combine multiple goals into a single scalar fitness function,the Non-Dominated Sorting Genetic Algorithm II(NSGA II) is explicitly designed for multi-objective optimisation tasks where objectives are fundamentally in conflict. In our context of rule extraction ,achieving high fidelity ,achieving high fidelity to the underlying Artificial Neural Network often comes at the expense of comprehensibility.

NSGA II addresses this trade-off by evolving a population of solutions toward a Pareto front which is a set of optimal solutions where no single objective can be improved without reducing the other.The algorithm operates on two core mechanisms :
1)Fast Non-Dominated Sorting: The population is ranked into hierarchical fronts based on pareto dominance .An individual rule set is considered “non-dominated” if no other rule set in the population is superior across all objectives.
2)Crowding Distance: To maintain diversity within the population and prevent premature convergence on a local optimum,NSGA II calculates a crowding distance for each individual.This metric estimates the density of solutions surrounding a particular rule set in the objective space ,prioritising individuals residing in less crowded regions during the selection phase
By utilising elitism ,which means carrying the best solutions from the combined parents and offspring populations into the next generation ,NSGA II ensures that high quality rule sets are not lost during evolution. For this RED XAI framework ,NSGA II is highly advantageous as it naturally outputs a diverse spectrum of rule sets ,allowing the PROMETHEE II multi criteria framework to later rank and select the optimal balance between mitigating “Rule Explosion” and maintaining statistical fidelity.

3.6 Auditing Quality and Robustness in XAI

3.6.1 Defining Quality in Symbolic Explanations: Fidelity, Comprehensibility, and Stability

When extracting "If-Then" rules from complex black box models, the quality of the rules are usually measured across three conflicting dimensions:

Fidelity - This metric measures the degree to which an explanation accurately mimics the behaviour of the black-box model (Guidotti, 2021). High fidelity translates as the extracted rules being highly faithful to the model’s actual decision-making and not falling into the disagreement problem (different explainability methods produce varying explanations for the same AI model)or just being an approximation (Miró-Nicolau et al., 2025).

Comprehensibility - Is defined by the sparsity of the model or the length or number of the generated rules for explanation (Arrieta et al.,2020; Yedjour et al., 2024). It relates to the cognitive load required for humans to understand and interpret the explanation. Literature suggests that as the number of conditions in a rule increases, human understanding decays, a phenomenon often referred to as Rule Explosion (Arrieta et al.,2020; Yedjour et al., 2024).

Stability refers to the consistency of an explanation when subjected to different subsets of training data or the inherent stochasticity of the training process [19]. Although stability has been studied primarily in image classification settings, Ye and Farnia (2025) establish that the underlying mechanism, explanation variance induced by model sensitivity to small perturbations, is a property of the model's loss landscape geometry rather than the data modality, and therefore generalises to tabular settings. In cancer genomics specifically, where batch effects and sample preparation variation introduce systematic noise into the feature space (Partin et al., 2023), an explanation that changes drastically due to negligible perturbations is considered fragile and clinically unreliable. This fragility not only erodes user trust (Wicker et al., 2022) but also hinders the clinician's ability to discover genuine, generalisable biological relationships, restricting explainability to debugging misclassifications rather than generating medical insight (Ye & Farnia, 2025).

3.6.2 Explanation Fragility and the Generalisation Gap

As established in Section 3.4.2, the Clever Hans effect represents a core threat to explanation reliability in overfitted models. The critical question for this project is how to detect and quantify this fragility before explanations are extracted. The Generalisation Gap, the difference between training and test accuracy (Keskar et al., 2016; Springer et al., 2024), serves as this primary diagnostic indicator, as models with a wide gap tend to overfit to training artefacts rather than learning genuine biological structure. The danger of this overfitting is hidden by traditional interpretability; Adebayo et al. (2018), through sanity checks, showed that popular XAI methods are independent of the model's parameters. This means that they can produce explanations that visually look convincing even for models that have actually just memorised noise. Consequently, this highlights the necessity of shifting toward a 'RED XAI' approach (Biecek & Samek, 2024), using XAI as an active diagnostic tool to interrogate the model and ensure it is learning grounded biological truths rather than mathematical coincidences resulting from reality drift (Longo et al., 2024).

3.6.3 Multi-Criteria Decision Analysis: PROMETHEE II

Given that fidelity, comprehensibility, and stability often exist in a state of trade-off (e.g., a highly faithful rule may be too complex to understand), a structured method for ranking rule quality is required (Yedjour et al., 2024; Ye & Farnia, 2025; Brans & Vincke, 1985).

PROMETHEE II (Preference Ranking Organisation Method for Enrichment Evaluations) is a Multi-Criteria Decision Analysis (MCDA) technique designed to rank alternatives based on multiple conflicting criteria (Yedjour et al., 2024; Matejová & Paralič, 2025). Unlike simple averaging, PROMETHEE II uses pairwise comparisons to establish an outranking relationship between different explanations, producing a complete ranking of all alternatives rather than only identifying a subset of optimal solutions as methods such as TOPSIS or VIKOR do (Yedjour et al., 2024; Matejová & Paralič, 2025). This complete ranking property is particularly important in the rule extraction context, where the goal is not merely to identify a single best rule set but to compare the full distribution of rule quality across the baseline and SAM-regularised models. Recent applications in rule extraction through the GPRO_RX system have demonstrated that PROMETHEE-assisted ranking can successfully identify optimal rules that maintain high data coverage while significantly reducing complexity (Yedjour et al., 2024). Following this precedent, and given that GPRO_RX represents the closest methodological antecedent to the evolutionary rule extraction pipeline used in this project, PROMETHEE II is adopted as the ranking framework here.

3.6.4 Biological Grounding: Pathway Enrichment (GSEA)

Finally, in the domain of cancer genomics, statistical fidelity is insufficient without biological verification. Gene Set Enrichment Analysis (GSEA) is a method used to determine whether a set of genes (such as those identified in an extracted rule) shows statistically significant, concordant differences between two

biological states (e.g., cancer vs. healthy) (Subramanian et al., 2005). By cross-referencing extracted rules with established databases like KEGG or Enrichr, researchers can verify if the AI's "logic" aligns with known molecular pathways (e.g., Apoptosis or DNA repair), thereby providing the ultimate "sanity check" for clinical relevance.

3.7 Critical Analysis

The literature reviewed in Section 3 reveals the existence of a diagnostic gap at the intersection of model robustness and explanation quality.

Section 3.1 established that XAI exists in two cultures. Blue XAI has dominated the field in terms of adoption, producing methods that justify model outputs to end users. Red XAI, as formalised by Biecek and Samek (2024), reorients XAI as an active interrogation tool, one that questions whether a model deserves to be explained at all before any explanation is surfaced. The RED XAI philosophy is well-articulated in theory; what remains is a complete pipeline that implements this philosophy from overfitting diagnosis through to measurable improvement in symbolic explanation quality. This project fills that gap.

The GDSC dataset, while invaluable as a pharmacogenomic resource, exhibits the P » N structure that reliably induces overfitting in unregularised deep models (Partin et al., 2023). Models must process tens of thousands of gene expression features measured across a comparatively small number of unique cancer cell lines, even though these are tested against hundreds of drugs to form a large dataset of combinations (Wu et al., 2025). Klauschen et al. (2024) and Partin et al. (2023) demonstrate that clinicians cannot verify whether a model's predictions are grounded in genuine oncogenic biology or learned shortcuts without an explicit auditing mechanism. The literature in cancer AI has produced increasingly powerful predictive models but has not systematically addressed the question of whether those models are ready to be explained (Wicker et al., 2022; Ye & Farnia, 2025). The distinction matters clinically: a biologically incoherent rule extracted from an overfitted model may appear statistically valid while pointing clinicians toward irrelevant or even harmful treatment decisions (Pahde et al., 2023; Klauschen et al., 2024).

Section 3.3 reviewed the theoretical relationship between loss landscape geometry and generalisation, establishing that models converging to sharp minima exhibit brittle, poorly generalising representations (Li et al., 2018; Keskar et al., 2016). SAM was reviewed as an intervention that explicitly seeks flat minima during training (Springer et al., 2024). What the literature has not established, however, is whether the geometric improvement that SAM produces in the loss landscape translates into measurable improvement in the quality of symbolic rules extracted from the model. The connection between landscape geometry and explanation quality has been theorised but not empirically demonstrated in the pharmacogenomic context. This project provides that empirical test.

Section 3.4 reviewed the ANN literature relevant to this project, covering the DeepCCDS architecture and the Clever Hans problem as documented by Pahde et al. (2023). The R2R framework represents the closest methodological precedent: it uses XAI to identify model artefacts, localises them, and corrects model behaviour iteratively. However, R2R operates on image data, uses LRP-based attribution heatmaps as its explanation medium, and corrects models through fine-tuning with artefact masks. It does not address the tabular genomic domain, does not use symbolic rule extraction as its explanation modality, and does not investigate whether regularisation applied at the optimisation stage can prevent the Clever Hans problem from arising rather than repairing it after the fact. This project is therefore differentiated from R2R in domain, explanation type, intervention mechanism, and causal direction.

Section 3.5 reviewed the evolutionary algorithm literature, specifically GAs and NSGA-II as applied to rule extraction. The GPRO_RX system demonstrated that multi-objective genetic algorithms guided by PROMETHEE-assisted ranking can produce rule sets that outperform single-objective extraction methods on the combined criteria of fidelity and comprehensibility (Yedjour et al., 2024). However, GPRO_RX and the broader EA rule extraction literature evaluate their methods on models with no deliberate control over the model's generalisation state, rule extraction is applied to whatever model happens to have been trained, without investigating whether the model's level of overfitting mediates rule quality. The comparative value of GA versus NSGA-II for rule extraction from models at different points on the generalisation spectrum remains unexplored. This project introduces that control by ex-

tracting rules from both a deliberately overfitted baseline and a SAM-regularised model, isolating the contribution of model health to rule quality independently of the extraction algorithm used.

Section 3.6 reviewed the quality framework for symbolic explanations, fidelity, comprehensibility, and stability as defined by Arrieta et al. (2020) and operationalised through PROMETHEE II ranking by Yedjour et al. (2024). Adebayo et al. (2018) showed through sanity checks that popular XAI methods can produce visually convincing explanations even for models that have memorised noise. To the best of our knowledge, the literature does not establish a quantitative relationship between the size of the generalisation gap and the degradation of rule quality metrics. Whether a smaller generalisation gap is a statistically reliable predictor of higher fidelity and lower rule complexity, the core hypothesis of this project, has not been tested. Section 3.6.4 further noted that even statistically valid rules require biological verification through pathway enrichment analysis to confirm clinical relevance, introducing a layer of grounding that purely metric-based XAI evaluation cannot provide.

Taken together, the literature has established four things independently: that overfitting produces unreliable explanations (Pahde et al., 2023; Adebayo et al., 2018); that loss landscape geometry is a measurable indicator of generalisation quality (Li et al., 2018); that SAM can improve landscape geometry (Foret et al., 2021); and that evolutionary algorithms can extract high-quality symbolic rules from neural networks when properly configured (Yedjour et al., 2024). The contribution of this project is not any single component, the ANN baseline, the SAM optimiser, the GA, the NSGA-II, and the PROMETHEE ranking are all individually established. The contribution is the controlled pipeline that connects them, the experimental structure that isolates model health as the independent variable, and the empirical evidence it generates about the relationship between generalisation and explainability in the pharmacogenomic domain.

While Yedjour et al. [20] demonstrated that PROMETHEE-assisted multi-objective genetic algorithm rule extraction outperforms single-objective methods, their evaluation was conducted exclusively on the UCI benchmark datasets and was not tested on high-dimensional genomic data, where the curse of dimensionality and the presence of batch effects present substantially different challenges to rule search spaces.

Springer et al. [16] established that SAM produces more balanced feature representations than conventional regularisers, but their analysis was limited to image classification benchmarks and does not provide evidence of this effect on tabular pharmacogenomic data or on the downstream quality of symbolic rule extraction.

Pahde et al. [13], while providing a compelling audit-and-fix framework through Reveal to Revise, operate exclusively in the image classification domain using LRP-based attribution heatmaps. Their correction mechanism is applied post hoc through fine-tuning with artefact masks, whereas this project intervenes at the optimisation stage to prevent Clever Hans behaviour from arising rather than correcting it after the fact.

Li et al. [11] provide strong theoretical and empirical evidence connecting flat minima to improved generalisation, but do not investigate how loss landscape geometry affects the quality of downstream symbolic explanations, leaving the connection between landscape smoothness and explanation fidelity as an empirically untested claim that this project is designed to address.

Adebayo et al. [1] demonstrated through sanity checks that popular attribution methods can produce convincing explanations for models that have memorised noise, but their study did not evaluate symbolic rule-based explanation methods or propose a regularisation intervention as a corrective mechanism, leaving the question of how to prevent rather than detect this behaviour unresolved.

3.7.1 Contributions

This project makes the following concrete contributions to the literature. The first is methodological. It introduces and evaluates a RED XAI pipeline that uses the generalisation gap as a diagnostic gatekeeper for symbolic explanation quality in cancer pharmacogenomics – a combination of domain, explanation type, and diagnostic mechanism that, to the best of our knowledge, has not been previously investigated.

The second is empirical. It provides a controlled experimental test of whether SAM regularisation, by smoothing the loss landscape, produces a statistically significant improvement in the fidelity and comprehensibility of evolutionary rule extraction on GDSC drug sensitivity data. Statistical rigour is ensured through Wilcoxon signed-rank testing across multiple experimental runs, following the precedent established by Yedjour et al. (2024), with the number of runs to be confirmed against the requirements of the adopted methodology in Section 4. [PLACEHOLDER – confirm run count- justificatio needed]

The third is evaluative. It provides a comparative assessment of two evolutionary rule extraction approaches – single-objective GA and multi-objective NSGA-II – applied to both a baseline overfitted model and a SAM-regularised model, across a consistent evaluation framework informed by Arrieta et al. (2020) and ranked through PROMETHEE II. This comparative structure generates evidence about which extraction method is more resilient to model fragility, a question that has not been addressed in the existing EA-XAI literature. The fourth is diagnostic. The loss landscape visualisation produced for both the baseline and SAM-regularised model constitutes a novel proposed indicator of explainability readiness – a geometric measure of whether a model is positioned to yield reliable explanations before any extraction is attempted. This moves the XAI quality question upstream, from the explanation evaluation stage to the model training stage, consistent with the RED XAI philosophy of interrogating the model rather than justifying it.

4 Methodology

4.1 Research Approach

This study adopts a *proof by demonstration* research methodology (NICOGSlides). The hypothesis is evaluated through a controlled experiment in which the optimisation strategy (Adam baseline versus SAM intervention) serves as the independent variable, and rule fidelity, comprehensibility, and stability serve as dependent variables. All other components of the pipeline — the DeepCCDS architecture, the GDSC dataset, the data splits, the EA configuration, and the evaluation framework — are held constant across both experimental conditions. This controlled substitution isolates the contribution of loss landscape geometry to explanation quality and ensures that observed differences in rule quality can be attributed specifically to the regularisation intervention rather than to confounding factors.

4.1.1 Technical Specifications

Experiments are implemented in Python [VERSION] using PyTorch [VERSION] for model training, the Distributed Evolutionary Algorithms in Python (DEAP) framework [VERSION] for GA and NSGA-II rule extraction, pymcdm [VERSION] for PROMETHEE II ranking, and GSEApy [VERSION] for pathway enrichment analysis. Training is conducted on [HARDWARE: e.g. NVIDIA GPU model or CPU specification]. The SAM optimiser approximately doubles per-epoch training time relative to the Adam baseline due to its two-step update procedure; this is accounted for in the project timeline in Section 5. All experiments are run on the fixed 80/10/10 train/validation/test split described in Section 4.1.3 to ensure reproducibility across all team members.

4.2 ANN: Baseline Training and Overfitting Diagnosis

4.3 Pipeline Overview

4.3.1 ANN: Baseline Training and Overfitting Diagnosis

Person 1 is responsible for training the baseline neural network on the preprocessed GDSC dataset and establishing the "Overfitting Baseline" – the starting point of the RED XAI diagnostic pipeline. This section details the model selection rationale, training configuration, and the specific criteria used to confirm that the baseline model exhibits the high Generalisation Gap required for Stage 1 of the experimental design.

4.3.2 Model Selection: DeepCCDS as the Shared Platform

DeepCCDS (Wu et al., 2025) is adopted as the shared neural network platform for the reasons detailed in Section 3.4.1. Critically for the experimental design, the same architecture can be trained under two distinct optimisation regimes – standard Adam without regularisation (Stage 1 Baseline) and SAM (Stage 2 Intervention) – allowing for a controlled comparison that isolates the effect of loss landscape geometry on explanation quality. The group's contribution is not the construction of a novel architecture from scratch. Rather, it is everything built around the network: the evolutionary rule extraction algorithms (Persons 2 and 3), the SAM regularisation intervention (Person 4), and the multi-criteria evaluation framework (Person 5). Person 1's role is to establish the controlled baseline and ensure a standardised model checkpoint is shared with all team members.

4.3.3 Dataset and Preprocessing Interface

The GDSC dataset, as preprocessed following the Alipour (2024) pipeline, provides the training data. The preprocessing steps – merging of the three GDSC sub-datasets (cell line features, drug information, and drug response measurements), removal of duplicates, handling of missing values, and encoding of categorical features – produce a structured tabular dataset where each row represents a cell line–drug combination characterised by gene expression features and a continuous drug sensitivity target (LN IC_{50}). Prior to model training, the following additional preprocessing steps are applied to ensure compatibility with DeepCCDS and the downstream rule extraction pipeline: The continuous gene expression features are Z-score normalised across all samples, such that each gene's expression is expressed relative to the population mean and standard deviation. This is consistent with the biological discretisation described in Section 4.1, which categorises features into High Expression ($Z > 1$), Low Expression ($Z < -1$), and Intermediate Expression for the symbolic rule extraction stage. The train/validation/test split is fixed at 80%/10%/10% and determined prior to any model training. This split is documented as a shared interface specification and used by all team members (Persons 1, 2, 3, and 5) without modification, ensuring that rule quality metrics are computed on identical held-out data throughout both stages of the experiment.

4.3.4 Stage 1: Training the Overfitting Baseline

The baseline model is trained using the Adam optimiser with a learning rate of 0.001 and no explicit regularisation (no weight decay, no dropout). To allow clear overfitting to emerge, training continues until the validation loss has not improved for a fixed number of consecutive epochs, a standard early stopping criterion applied in reverse, where the patience parameter is used to confirm that the generalisation gap has stabilised rather than to halt training prematurely (Goodfellow et al., 2016). The stopping point is defined as the epoch at which training loss continues to decrease while validation loss has plateaued or begun to increase for at least [PLACEHOLDER confirm patience parameter] consecutive epochs, producing a measurable and stable generalisation gap. The Generalisation Gap is formally defined as:
Generalisation Gap = Training Accuracy - Validation Accuracy
or equivalently in terms of loss:
Generalisation Gap = Validation Loss - Training Loss
The magnitude of the gap is recorded in terms of both R^2 and RMSE on the regression target (LN IC_{50}) and constitutes the confirmed Fragility Baseline. This mirrors the diagnostic framing established by Pahde et al. (2023), where a model achieving high training performance through spurious correlations serves as the starting condition for the bias identification and correction cycle. [PLACEHOLDER if a domain-specific threshold can be identified from Partin et al. (2023) or Wu et al. (2025), insert here as a reference benchmark] The trained baseline model weights are saved as a checkpoint and shared with Persons 2, 3, and 5. Persons 2 and 3 load this checkpoint and do not retrain, ensuring that their evolutionary rule extraction algorithms operate on exactly the same model in both Stage 1 and Stage 2.

4.3.5 Confirming the Clever Hans Baseline

In addition to the generalisation gap metric, a qualitative sanity check is applied, inspired by the approach of Pahde et al. (2023): feature attribution maps are generated for a sample of training and validation predictions using a basic post-hoc attribution method such as input gradients or a lightweight LRP implementation. If the baseline model is genuinely overfitting to training noise, attribution maps for correctly predicted training samples will exhibit diffuse or biologically incoherent relevance patterns,

assigning high importance to genes with no established role in the relevant oncogenic pathway, while validation attribution patterns will be inconsistent across samples. This qualitative check documents the fragility of the baseline model's explanations and motivates the subsequent SAM intervention. The generalisation gap measurement and attribution patterns together constitute the quantitative and qualitative inputs to the Stage 1 audit, formally establishing the Fragility Baseline scorecard against which Stage 2 results are compared.

4.4 Genetic Algorithms for Rule Extraction

Rule extraction from the trained baseline model is performed using a Genetic Algorithm implemented via the Distributed Evolutionary Algorithms in Python (DEAP) framework. Following the RED XAI philosophy established in Section 3.1.1, the GA is used not merely to produce interpretable if-then rules but as a diagnostic tool to expose the decision logic of the overfitted model and evaluate its coherence. The extraction process balances fidelity and comprehensibility as competing objectives throughout (Yedjour et al., 2024).

4.4.1 Preparation for Rule Extraction

Prior to rule extraction, loss landscape visualisation is applied to the trained baseline model to characterise the sharpness of the minima to which it has converged and to document the overfitting behaviour that motivates the SAM intervention in Stage 2. This visualisation serves as a qualitative complement to the generalisation gap metric established in Section 4.1.4 and provides a geometric baseline against which the SAM-regularised model's landscape is later compared (Li et al., 2018).

4.4.2 Pre-processing Data for Rule Extraction

Rather than operating on continuous inputs, rule extraction requires discretisation of the gene expression features. Continuous input ranges are divided into k partitions of equal length, producing discrete categories such as high expression, intermediate expression, and low expression as described in Section 4.1.3. Rules are then extracted in the format: if X_1 is in partition A and X_2 is in partition B, then predicted response Y, where each condition corresponds to a discretised feature partition (Yedjour et al., 2024).

4.4.3 Implementation of GA via DEAP

An evolutionary loop is configured within the DEAP framework to discover optimal rule sets. Each individual chromosome in the population represents a candidate rule set encoded as a vector, where each element corresponds to a discretised attribute partition or a special value of , 1 to indicate that the attribute is not involved in the rule (Yedjour et al., 2024). The fitness function adopts a multi-objective formulation, simultaneously optimising fidelity to the black-box model, predictive accuracy on held-out data, and comprehensibility measured by rule length (Yedjour et al., 2024). Tournament selection is used to identify parent individuals for reproduction. Two-point crossover combines rule segments from two parent chromosomes to produce offspring, and mutation introduces random changes – such as adding or removing a condition – to maintain population diversity and prevent premature convergence on local optima (Yedjour et al., 2024).

4.4.4 Rule Pruning and Population Diversity

Following each generation, rule pruning is applied to remove conditions with low confidence and support, maintaining a concise rule set and reducing rule explosion. Aging evolution is additionally applied as a diversity mechanism to prevent the population from converging prematurely on local optima, ensuring that the evolutionary search continues to explore the rule space rather than stagnating on a suboptimal solution (Yedjour et al., 2024).

4.4.5 Validation and Execution

The GA is executed for 50 generations with a population size of 35 to 45 individuals, following the configuration established by Yedjour et al. (2024). The output of each run is a Pareto front representing a

set of non-dominated rule sets that each reflect a different trade-off between fidelity and comprehensibility. The final selected rule set is the one that achieves the best balance across these objectives as determined by PROMETHEE II ranking in Section 4.5.1.

4.5 NSGA-II for Rule Extraction

NSGA-II is implemented as a comparative multi-objective rule extraction technique alongside the GA, providing the evaluative structure needed to determine which evolutionary algorithm produces more resilient rules under conditions of model fragility. The implementation uses the same DEAP framework and operates on the identical discretised GDSC dataset and saved model checkpoints used in the GA pipeline, ensuring experimental consistency across both extraction methods.

Unlike the GA, which combines competing criteria into a composite fitness function, NSGA-II strictly separates the objectives and evaluates them simultaneously. The algorithm maximises fidelity and accuracy, measured by the F1 score of the extracted rules against the predictions of the underlying DeepCCDS model, while minimising rule length, measured by the total number of conditions or premises per rule, to enforce comprehensibility and prevent rule explosion (Yedjour et al., 2024).

Selection uses a binary tournament operator based on the crowded comparison approach: when comparing two rule sets, the algorithm prefers the one with the lower non-domination rank, and where both belong to the same front, selects the rule set with the larger crowding distance to maintain genetic diversity. Two-point crossover and random mutation are applied identically to the GA configuration to ensure comparability across both extraction methods (Yedjour et al., 2024).

The NSGA-II loop runs for the same number of generations and population size as the GA. Unlike the GA, the final output is not a single best rule set but a Pareto front of non-dominated rule sets, which is passed directly to the PROMETHEE II ranking framework in Section 4.5.1 for both Stage 1 and Stage 2 evaluation.

4.6 SAM Regularisation

The second stage of the experimental pipeline retrains the DeepCCDS architecture using the Sharpness-Aware Minimisation (SAM) optimiser to reduce the generalisation gap and improve loss landscape smoothness.

4.6.1 Intervention Objective

SAM is applied to the same DeepCCDS architecture, dataset splits, preprocessing pipeline, and training configuration used in the baseline model, ensuring that the optimisation strategy is the sole variable distinguishing the two models. This controlled substitution isolates the effect of loss landscape geometry on three downstream outcomes: the generalisation gap, the smoothness of the loss landscape as measured through visualisation, and the quality of rules extracted by the evolutionary algorithms in Sections 4.2 and 4.3.

4.6.2 SAM Optimisation Mechanism

SAM modifies the standard training objective by introducing a neighbourhood-based perturbation step. Rather than minimising the loss at the current parameters θ alone, the optimiser evaluates the worst-case loss within a local ϵ-ball around the current weights, effectively minimising the loss at perturbed parameters $\theta + \epsilon$. This forces the model to favour solutions that remain stable under small parameter perturbations, penalising sharp curvature in the loss surface and encouraging convergence toward flat minima (Foret et al., 2021; Li et al., 2018).

4.6.3 Expected Effect on Generalisation

The baseline model is characterised by high training accuracy, lower validation performance, and a large generalisation gap, indicating that the model has memorised training-specific noise rather than learning stable biological patterns. By converging on flatter minima, the SAM-regularised model is expected

to reduce the training-validation discrepancy, producing more balanced performance across both sets. Following Springer et al. (2024), this improvement in generalisation is expected to produce more balanced feature representations, which in turn should yield extracted rules with fewer spurious conditions and higher biological coherence than those produced by the baseline model.

4.7 Evaluation Strategy

The evaluation strategy employs a Multi-Criteria Decision Analysis (MCDA) framework to objectively rank the extracted rules, moving beyond a single-metric (accuracy-based) approach (Matejová & Paralič, 2025).

4.7.1 Multi-Criteria Ranking via PROMETHEE II

To determine the top 10 rules from the thousands generated by the genetic algorithm, the PROMETHEE II method will be implemented using the pymcdm library (Yedjour et al., 2024). Each rule is treated as an alternative and ranked based on three conflicting criteria:
Fidelity: Measured using the F1-score of the rule's predictions compared to the black-box model's predictions. This ensures that the rule acts as a faithful surrogate (Guidotti, 2021).
Comprehensibility: Measured by the Length of the Rule (i.e.,number of conditions). Shorter rules are prioritised to avoid rule explosion (Yedjour et al., 2024).
Stability: Calculated by the variance in rule performance across different cancer types in the GDSC dataset (Ye and Farnia, 2025).

4.7.2 Statistical Significance (Wilcoxon Signed-Rank Test)

To confirm that improvements in rule quality between the baseline and SAM-regularised models are statistically significant rather than coincidental, a Wilcoxon signed-rank test is applied to the distribution of net outranking flows produced by PROMETHEE II for the top 10 rules of each model. A non-parametric test is required because the distribution of rule quality metrics such as outranking flows cannot be assumed to follow a normal distribution. The Wilcoxon signed-rank test is adopted following the precedent of Yedjour et al. (2024), who apply it to compare PROMETHEE-scored rule sets across extraction conditions, with a p-value threshold of 0.05 used to determine statistical significance. A result below this threshold is interpreted as evidence that the difference in rule quality between the baseline and SAM conditions is unlikely to be due to random chance.

4.7.3 Automated Biological Verification (GSEApy)

The genes identified in the top-ranked rules from both the baseline and SAM-regularised models are submitted to over-representation analysis using the GSEApy library (Fang et al., 2023), which provides a programmatic interface to the Enrichr and KEGG pathway databases. For each rule set, GSEApy tests whether the genes appearing in the extracted rules are statistically enriched in known biological pathways relative to a background gene set. The pathway enrichment p-value is used to determine whether the rules align with established oncogenic mechanisms, such as apoptosis or DNA repair pathways, or whether they represent statistically arbitrary associations with no biological grounding (Subramanian et al., 2005). A significantly higher enrichment score in the SAM-regularised rule set relative to the baseline would constitute evidence that improved generalisation produces not only statistically better rules but biologically more meaningful ones, directly addressing sub-research question 4 from Section 2.

4.7.4 SOTA Baseline Comparison: Decision Tree Rule Extraction

To contextualise the quality of rules produced by the GA and NSGA-II pipelines, a CART decision tree [7] is included as a deterministic rule extraction baseline, representing the most widely adopted alternative to evolutionary rule extraction in the XAI literature [3]. The decision tree is fitted to the predictions of both the baseline and SAM-regularised DeepCCDS models using scikit-learn, with maximum depth constrained to prevent rule explosion. Rules are extracted from the fitted tree and evaluated under the identical PROMETHEE II framework described in Section 4.5.1, using the same fidelity, comprehensibility, and stability criteria. This enables a direct comparison between deterministic surrogate-based rule

extraction and the proposed evolutionary approach across both model conditions. The Wilcoxon signed-rank test described in Section 4.5.2 is applied to the PROMETHEE outranking flow distributions of the decision tree rules and the GA/NSGA-II rules to determine whether the improvement from evolutionary extraction is statistically significant, and whether this advantage is maintained after SAM regularisation.

4.8 Experiments

The research follows a two-stage experimental design to isolate the impact of model sharpness on explanation quality. In Stage 1, the Overfitting Baseline is established as the RED XAI diagnostic starting point. DeepCCDS is trained using the Adam optimiser until a stable generalisation gap has emerged using the early stopping criterion defined in Section 4.1.4, confirming the Clever Hans baseline. Rules are then extracted from this overfitted model by both the GA and NSGA-II pipelines. The resulting rule sets are audited using PROMETHEE II ranking and GSEA pathway enrichment analysis to document the Fragility Baseline scorecard. In Stage 2, the Regularisation Intervention is applied. The same DeepCCDS architecture is retrained using the SAM optimiser to smooth the loss landscape and reduce the generalisation gap, as described in Section 4.4. Rules are re-extracted using identical GA and NSGA-II parameters to ensure comparability. The same PROMETHEE II and GSEA audit is performed to generate a second evaluation scorecard. Wilcoxon signed-rank tests are then applied to the distributions of PROMETHEE outranking flows from both stages to determine whether the improvement in rule quality is statistically significant.

Robustness Validation: Cancer Cell Line Encyclopedia (CCLE) As a stretch objective, subject to timeline constraints, the complete RED XAI pipeline will be applied to a second pharmacogenomic dataset: the Cancer Cell Line Encyclopedia (CCLE) [5]. The CCLE provides gene expression profiles and drug response data for cancer cell lines using a structurally compatible format to GDSC, and has been used alongside GDSC in several of the studies cited in this proposal [4, 14]. Applying the pipeline to CCLE data, with LN IC_{50} as the response target, would enable assessment of whether the observed advantage of SAM regularisation over the Adam baseline in terms of rule fidelity, comprehensibility, and biological validity generalises beyond the GDSC dataset. Consistent results across both datasets would substantially strengthen the robustness of the study's conclusions. No new statistical machinery is required; the same Wilcoxon signed-rank test and PROMETHEE II framework are applied, and results are reported comparatively.

4.9 Expected Results

Four primary outcomes are anticipated from this methodology. First, an inverse correlation is expected between the generalisation gap and rule comprehensibility: the SAM-regularised model should produce rules with fewer conditions than the baseline while maintaining comparable fidelity, consistent with the hypothesis that flatter minima yield more balanced feature representations (Springer et al., 2024).

Second, the rules extracted from the SAM-regularised model are expected to show higher stability scores across different cancer subtypes in the GDSC dataset, reflecting the capture of genuine biological signal rather than training-specific noise.

Third, the genes appearing in SAM-optimised rules are expected to show statistically significant enrichment in established KEGG pathways ($p < 0.05$) compared to baseline rules, providing biological grounding for the claim that regularisation improves not only statistical but clinical rule quality.

Fourth, and contingent on the preceding outcomes, the study aims to demonstrate that the generalisation gap serves as a reliable diagnostic indicator of explanation fragility, and that SAM regularisation constitutes an effective intervention for improving the fidelity, comprehensibility, and biological validity of symbolically extracted rules in cancer pharmacogenomics.

5 Planning

The project is structured across six months from June to November 2026. All five team members begin with shared boilerplate and pilot testing in June to establish a common codebase and verify the pipeline end-to-end. Data preparation and baseline model training are completed in July. Rule extraction by Persons 2 and 3, alongside the SAM intervention by Person 4, run in parallel across August and September. Person 5 conducts PROMETHEE II ranking, statistical validation, and biological verification on a rolling basis from August through October as rule sets are produced. Final evaluation, comparative analysis, and report writing are completed across October and November by all team members.

References

[1] J. Adebayo et al. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

[2] C. J. Anders et al. Finding and removing clever Hans: Using explanation methods to debug and improve deep models. *Information Fusion*, 77:261–295, 2022.

[3] A. B. Arrieta et al. Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion*, 58:82–115, 2020.

[4] D. Baptista, P. G. Ferreira, and M. Rocha. Deep learning for drug response prediction in cancer. *Briefings in Bioinformatics*, 22(1):360–379, 2021.

[5] J. Barretina et al. The cancer cell line encyclopedia enables predictive modelling of anticancer drug sensitivity. *Nature*, 483:603–607, 2012.

[6] Przemyslaw Biecek and Wojciech Samek. Position: Explain to question not to justify. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.

[7] L. Breiman, J. Friedman, C. J. Stone, and R. A. Olshen. *Classification and Regression Trees*. CRC Press, 1984.

[8] A. K. Dombrowski et al. Explanations can be manipulated and geometry is to blame. In *Advances in Neural Information Processing Systems*, 2022.

[9] P. Foret, A. Kleiner, H. Mobahi, and B. Neyshabur. Sharpness-aware minimization for efficiently improving generalization. In *International Conference on Learning Representations (ICLR)*, 2021.

[10] F. Klauschen et al. Toward explainable artificial intelligence for precision pathology. *Annual Review of Pathology: Mechanisms of Disease*, 19:541–570, 2024.

[11] H. Li, Z. Xu, G. Taylor, C. Studer, and T. Goldstein. Visualizing the loss landscape of neural nets. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

[12] L. Longo et al. Explainable artificial intelligence (XAI) 2.0: A manifesto of open challenges and interdisciplinary research directions. *Information Fusion*, 106:102301, 2024.

[13] F. Pahde, M. Dreyer, W. Samek, and S. Lapuschkin. Reveal to revise: An explainable AI life cycle for iterative bias correction of deep models, 2023.

[14] A. Partin et al. Deep learning methods for drug response prediction in cancer: Predominant and emerging trends. *Frontiers in Medicine*, 10:1086097, 2023.

[15] W. Samek et al. Evaluating the visualization of what a deep neural network has learned. *IEEE Transactions on Neural Networks and Learning Systems*, 28(11):2660–2673, 2017.

[16] J. M. Springer, V. Nagarajan, and A. Raghunathan. Sharpness-aware minimization enhances feature quality via balanced learning. In *International Conference on Learning Representations (ICLR)*, 2024.

[17] Y. Wu et al. DeepCCDS: Interpretable deep learning framework for predicting cancer cell drug sensitivity through characterizing cancer driver signals. *Advanced Science*, 2025.

[18] W. Yang et al. Genomics of drug sensitivity in cancer (GDSC): a resource for therapeutic biomarker discovery in cancer cells. *Nucleic Acids Research*, 41(D1):D955–D961, 2013.

[19] Z. Ye and K. Farnia. Gaussian smoothing in saliency maps: The stability-fidelity trade-off in neural network interpretability, 2024.

[20] H. Yedjour, D. Yedjour, and A. Benyettou. Rule extraction based on PROMETHEE-assisted multi-objective genetic algorithm for generating interpretable neural networks. *Applied Soft Computing*, page 111165, 2024.